AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2014

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Briefly describe how the knowledge of biology can improve the yield of the fishing industry. (3 Marks)

b. How is turgor important to plants? (2 Marks)

Q.2. (Total 4 Marks)

It is said, '*Papaya contains an enzyme which digests proteins.*'

Write a set of instructions (procedure) to test the given hypothesis. Your answer should include a 'control set-up'.

Q.3. (Total 3 Marks)

What may happen if G1 phase of the interphase of cell cycle does not take place?

Q.4. (Total 2 Marks)

Name any TWO intracellular enzymes in our body.

Q.5. (Total 6 Marks)

a. With the help of a labelled graph ONLY show the effect of light intensity on the rate of photosynthesis. (3 Marks)

Space for graph

b. The table shows the composition of diet of different vulnerable groups. (3 Marks)

Group	Diet
Child (4-6 years)	Meat, egg, rice and potato
Adult (above 50)	Fish, apple, bread and spinach
Labourer	

i. Why are the dietary requirements of a child different from those of an adult?

ii. Complete the given table by inserting the dietary requirements of a labourer.

Q.6. (Total 5 Marks)

a. Give an account of the effect of increased humidity on the rate of transpiration. (2 Marks)

b. State the causes of leukaemia and thalassemia. Also mention the treatment for these diseases. (3 Marks)

Q.7. (Total 7 Marks)

EITHER

a. Define biodiversity. Describe the basis of classification of living organisms with reference to homology, analogy and embryology.

OR

b. Given are names of some renowned biologists.

Robert Brown	Schleiden and Schwann	Aristotle
Robert Hooke	Antonie van Leewenhoek	

Following the progressive development of our knowledge of the cell from the earliest to the recent, describe the role of each scientist.

Q.8. (Total 8 Marks)

EITHER

a. Referring to the structure of a villus, describe the absorption of food in the small intestine. Your answer should also include the role of capillaries and lacteal.

OR

b. Highlighting the structure of phloem vessels describe how food is translocated in plants. Refer to Pressure Flow Mechanism in your answer.

END OF PAPER